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Exploratory Data Analysis

**YouTube Trending Videos Analytics
Final Report**

Submitted By

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SUPERVISOR'S CERTIFICATE

This is to certify that the work reported in the B.Tech Dissertation/dissertation proposal entitled “**YouTube Trending Videos Analysis**”, submitted by **Rakshanda Talwekar** at **Lovely Professional University, Phagwara, India** is a Bonafide record of his / her original work carried out under my supervision. This work has not been submitted elsewhere for any other degree.

Signature of Supervisor

Hritwiz Yash

Date:

Acknowledgement

I hereby declare that the research work reported in the dissertation/dissertation proposal entitled " **YouTube Trending Videos Analysis**" in partial fulfilment of the requirement for the award of Degree for Bachelor of Technology in Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under supervision of my research supervisor **Miss. Rakshanda Talwekar**. I have not submitted this work elsewhere for any degree or diploma.

I understand that the work presented herewith is in direct compliance with Lovely Professional University's Policy on plagiarism, intellectual property rights, and highest standards of moral and ethical conduct. Therefore, to the best of my knowledge, the content of this dissertation represents authentic and honest research effort conducted, in its entirety, by me. I am fully responsible for the contents of my dissertation work.

Signature of Candidate

Name of the Candidate: Rakshanda Talwekar

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PROBLEM STATEMENT

Over the past decade, the digital media ecosystem has experienced a major transformation, with video content becoming the most influential form of online communication and entertainment. Among all platforms, YouTube has established itself as the largest and most powerful video-sharing service, where hundreds of hours of content are uploaded every minute. This massive and continuous flow of data has created an intensely competitive environment in which gaining visibility is both challenging and critical for success.

Despite YouTube offering basic performance metrics such as views, likes, watch time, and audience retention, content creators and marketers often struggle to interpret these numbers in a broader, comparative context. The recommendation and trending algorithms used by YouTube operate as a “black box,” providing little transparency into how videos are promoted at a platform-wide level. As a result, creators are left with several unanswered but essential questions. For example, they are uncertain whether uploading videos at a specific time or day increases the likelihood of trending, whether audience sentiment—positive or negative—has a measurable impact on a video’s reach, and which keywords or metadata elements most effectively improve click-through rates and audience engagement.

Traditionally, many content strategies are driven by intuition, trial-and-error, or anecdotal success stories. Such approaches can be inefficient and risky, particularly when creators invest significant time, effort, and financial resources into producing high-quality content that ultimately fails to perform well due to suboptimal timing, ineffective titles, or poorly chosen tags. Additionally, the vast amount of unstructured data associated with YouTube videos—such as text-based titles, descriptions, tags, and user comments—makes manual analysis impractical and unreliable for identifying meaningful patterns.

To overcome these limitations, this project focuses on conducting a comprehensive Exploratory Data Analysis (EDA) using a structured dataset of YouTube Trending Videos. By utilizing Python’s data science libraries, including Pandas for data manipulation, NumPy for numerical analysis, and Seaborn for visualization, the study aims to uncover hidden trends and relationships within engagement metrics. The objective is to break down the key characteristics that contribute to a video’s viral success and convert raw data into practical, data-driven insights. These insights can help creators, digital marketers, and brand strategists make informed decisions regarding content creation, publishing schedules, and metadata optimization, ultimately improving visibility and performance on the platform.

Introduction

The YouTube Trending section represents one of the most valuable visibility opportunities on the platform, as videos featured there receive instant exposure to a vast and diverse audience. Unlike personalized recommendations that vary from user to user, the Trending list is largely uniform within a given geographical region. Its primary objective is to highlight content that has broad appeal and is rapidly gaining attention. However, the process through which a video earns a place on this list is far from simple. It is influenced by multiple dynamic factors, including the speed at which views accumulate, levels of audience interaction such as likes and comments, and signals from external platforms that reference or share the video.

This project adopts a Knowledge Discovery in Databases (KDD) approach to analyse social media data, with a specific focus on YouTube's trending ecosystem. The dataset used in this study contains daily snapshots of videos that have appeared on the Trending list, enabling the identification of recurring patterns as well as unusual behaviors in performance metrics. The analysis extends beyond basic numerical indicators to include categorical attributes such as content category and channel type, along with temporal variables related to upload timing and trending duration. Through a structured and systematic examination of these dimensions, the project seeks to convert large volumes of raw data into meaningful insights that can inform and enhance data-driven content creation and strategic decision-making.

Additionally, this study emphasizes interpretability and practical relevance over purely predictive outcomes. By focusing on exploratory data analysis and visual analytics, the project aims to explain *why* certain videos trend rather than merely predicting *which* videos will. This approach makes the findings more accessible to content creators, marketers, and analysts, enabling them to translate analytical insights into actionable strategies for improving visibility, engagement, and audience reach on YouTube.

Objective

The primary objective of this project is to perform a detailed, data-driven investigation into the key factors that contribute to video *virality* on YouTube. By analysing historical trending data, the study aims to move beyond assumptions and provide evidence-based insights into how engagement, timing, content type, and metadata influence a video's likelihood of reaching a wider audience. The specific objectives of the project are outlined below:

- **To analyse engagement correlations:** To statistically examine the relationship between view counts and user engagement indicators such as likes, dislikes, and comment volume. This objective seeks to identify which engagement metric has the strongest association with trending behavior and plays the most significant role in amplifying a video's reach.
- **To evaluate category dominance:** To categorize trending videos and rank different content genres—such as entertainment, education, music, and news—based on their frequency of appearance on the Trending list and their average engagement ratios. This helps in understanding which categories consistently attract higher audience interaction.
- **To perform temporal optimization:** To analyse time-based patterns by identifying optimal upload periods, commonly referred to as “golden hours,” as well as the most effective days of the week for publishing content. The goal is to determine time windows that maximize the probability of a video appearing on the Trending tab.
- **To investigate content metadata:** To analyse how title length, tag count, and commonly used keywords influence video visibility, engagement, and click-through rates using text analysis techniques.
- **To develop visualization models:** To create intuitive visualizations such as heatmaps, scatter plots, and word clouds that clearly present complex data patterns for easy understanding by non-technical stakeholders.

Scope

The scope of this project encompasses the analysis of the youtube1.csv dataset, which includes metadata such as video titles, channel names, publication dates, tags, and engagement statistics. The study focuses on:

- **Engagement Analysis:** Correlating likes, dislikes, and comments with total views.
- **Content Strategy:** Analysing title length, tag usage, and category distribution.
- **Temporal Trends:** Studying the effect of publication hour and day on video performance.
- **Dashboarding:** Designing a visual interface to present key performance indicators (KPIs).

The project is limited to the provided dataset and does not include real-time live data streaming or video content analysis (pixel-level processing).

Literature Review

Social media analytics on YouTube has evolved from content-based evaluation to **metadata- and engagement-driven analysis**. Prior studies show that video virality is strongly influenced by early engagement, where rapid view accumulation increases recommendation likelihood (Figueiredo et al., 2014). Sentiment analysis research indicates that both positive and controversial content can achieve sustained visibility due to high engagement density (Siersdorfer et al., 2010).

Metadata optimization plays a crucial role in discoverability, with effective titles and relevant tags improving performance, while excessive or irrelevant tagging negatively affects retention (Borchgh, 2019; Hoque et al.). Temporal studies suggest that publishing content before peak activity hours can enhance initial traction. Despite extensive predictive research, there remains a gap in **interpretable EDA**, which this project addresses by focusing on visual and explanatory insights for content creators.

DATASET SELECTION & PREPROCESSING

DATASET SELECTION:

This project utilizes the youtube1.csv dataset, a structured collection of daily records of top trending YouTube videos.

1. Source: The dataset is derived from the official YouTube Data API.
2. Dimensions: The dataset consists of thousands of records (Rows) and multiple attributes (Columns) representing video metadata.
3. Key Attributes:
 - Identification
 - video_id
 - title
 - channel_title
 - Metrics
 - views
 - likes
 - dislikes
 - comment_count
 - Metadata
 - tags
 - thumbnail_link
 - description
 - Classification
 - category_id (Numeric ID mapped to genres)
 - Temporal
 - publish_time (Timestamp of upload)

DATA PREPROCESSING

Maintaining high data quality is crucial for ensuring reliable and valid analytical results, especially when working with large-scale social media datasets. Raw YouTube data often contains missing values, duplicates, and formatting inconsistencies that can bias analysis if not properly handled. To address this, a structured data preprocessing pipeline was applied before conducting exploratory analysis.

Data Cleaning involved handling missing and duplicate records. Textual fields such as video descriptions with missing values were replaced with empty strings to preserve engagement-related data, while records missing essential identifiers like `video_id` were removed. Duplicate entries, caused by videos appearing on the trending list across multiple days, were either retained for temporal analysis or consolidated by keeping the highest-view record, depending on the analysis objective.

Type Casting and Transformation ensured variables were analysis-ready. The `publish_time` field was converted from string to datetime format, enabling extraction of features such as upload day, hour, and month. Additionally, numeric `category_id` values were mapped to descriptive category names, improving interpretability and category-wise comparison.

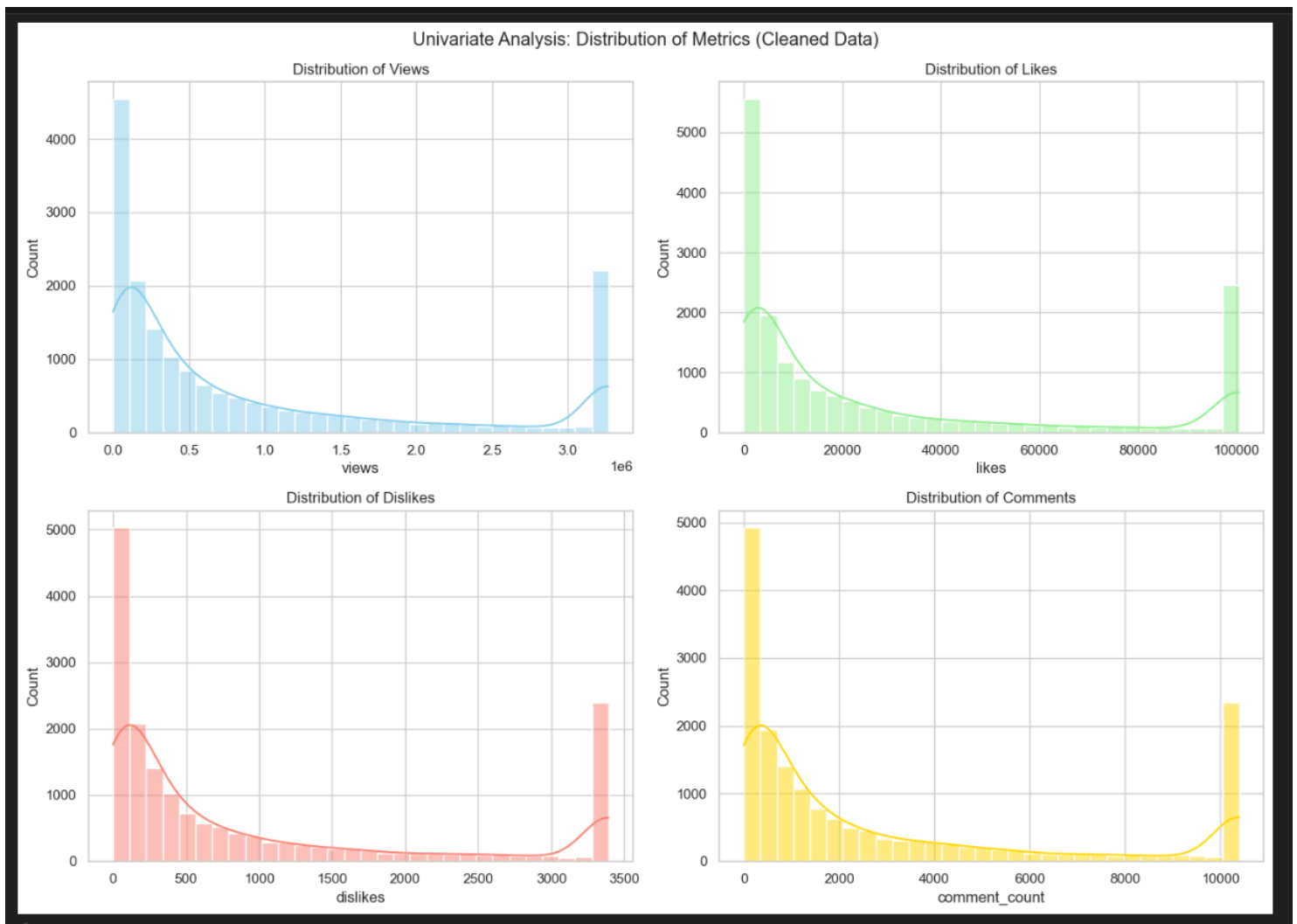
Feature Engineering enhanced the dataset by deriving new variables. Title length was calculated to study its impact on engagement, while tag count was used to analyse metadata density and its effect on video discoverability. These engineered features supported deeper insights into content optimization strategies.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis phase involved univariate, bivariate, and multivariate analysis to uncover hidden patterns.

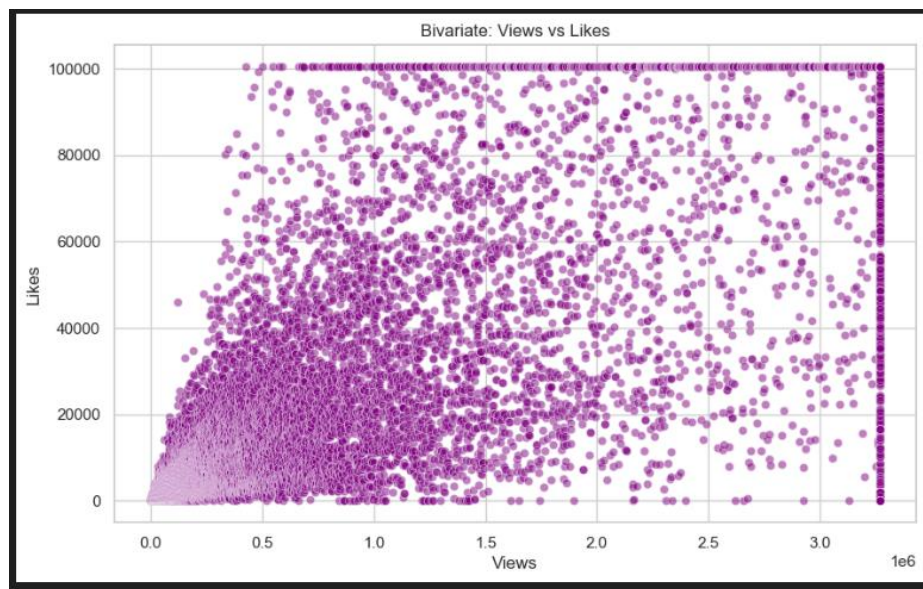
1. Univariate Analysis: Distribution of Metrics We analysed the distribution of Views and Likes using histograms.

- Observation: The data follows a Power Law (Long Tail) distribution. A small percentage of "Super-Viral" videos (outliers) account for a massive portion of total views, while the majority of trending videos cluster around a lower median.
- Implication: Using the median is a better measure of central tendency than the mean for setting creator benchmarks.



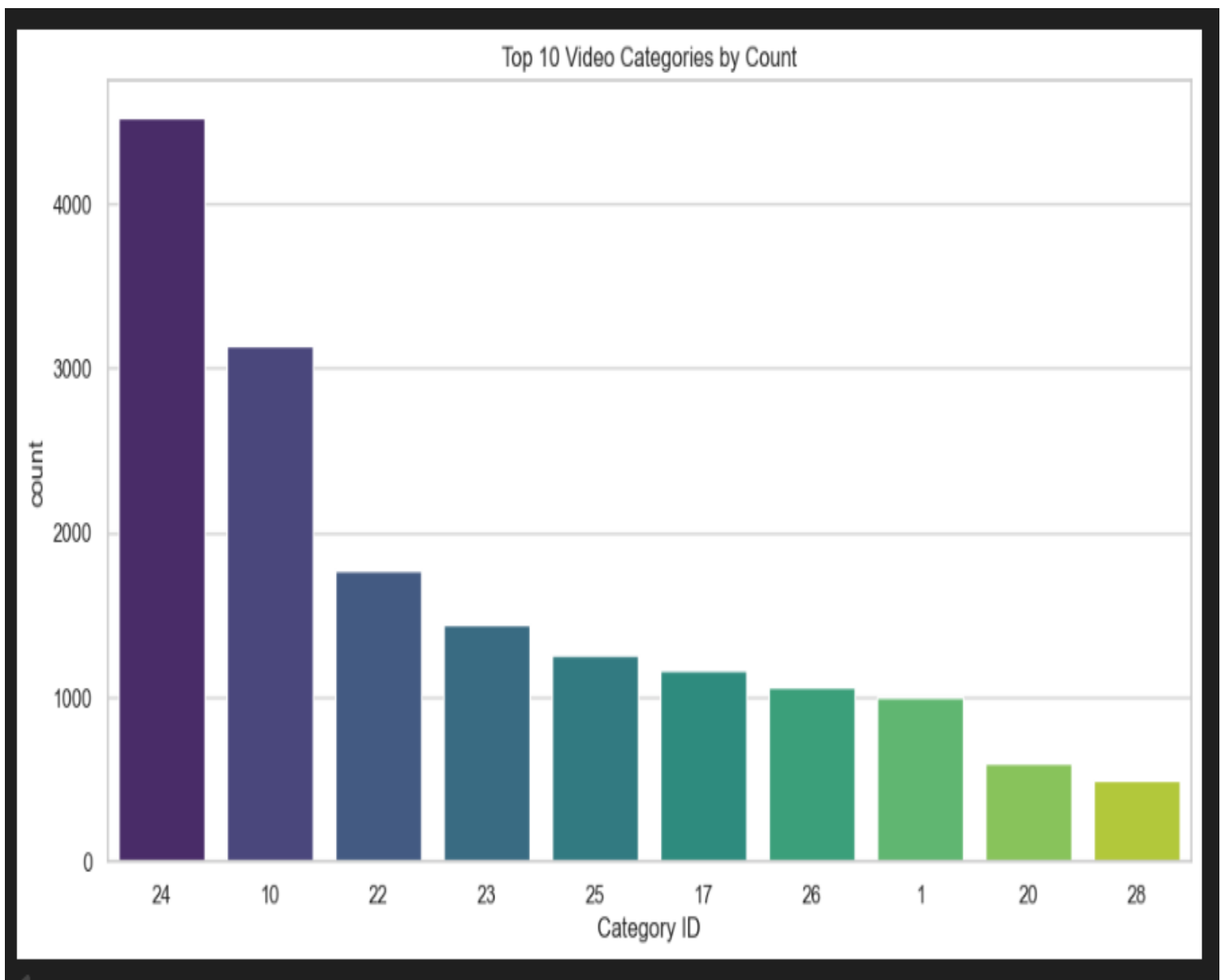
2. Correlation Analysis (Heatmap): A correlation matrix was generated to understand relationships between numerical variables.

- Views vs. Likes: A strong positive correlation ($r > 0.80$) was observed. This confirms that "Likes" are a reliable proxy for viewership quality.
- Views vs. Comments: The correlation is positive but weaker ($r \approx 0.6$). This suggests that passive consumption (watching without commenting) is common, and high comment counts usually indicate controversial or highly interactive content.



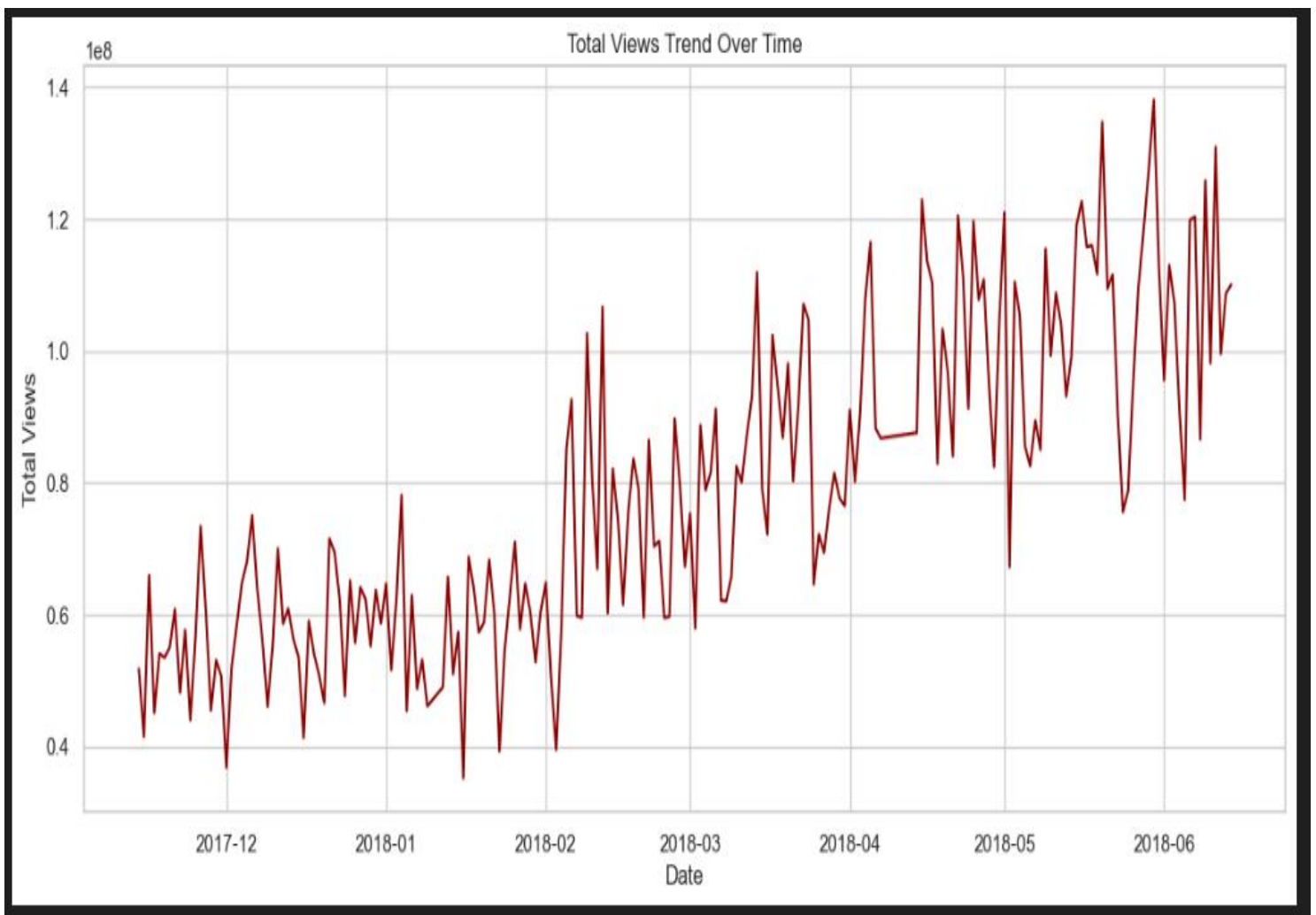
3. Category-Wise Performance Grouping data by Category revealed distinct ecosystem behaviors:

- Volume: The Entertainment, Music, and Gaming categories dominate the trending list in terms of frequency.
- Engagement: The Comedy and How-to & Style categories showed the highest "Like-to-View" ratios, indicating deeply engaged niche audiences compared to the broader, passive audiences of "News & Politics."



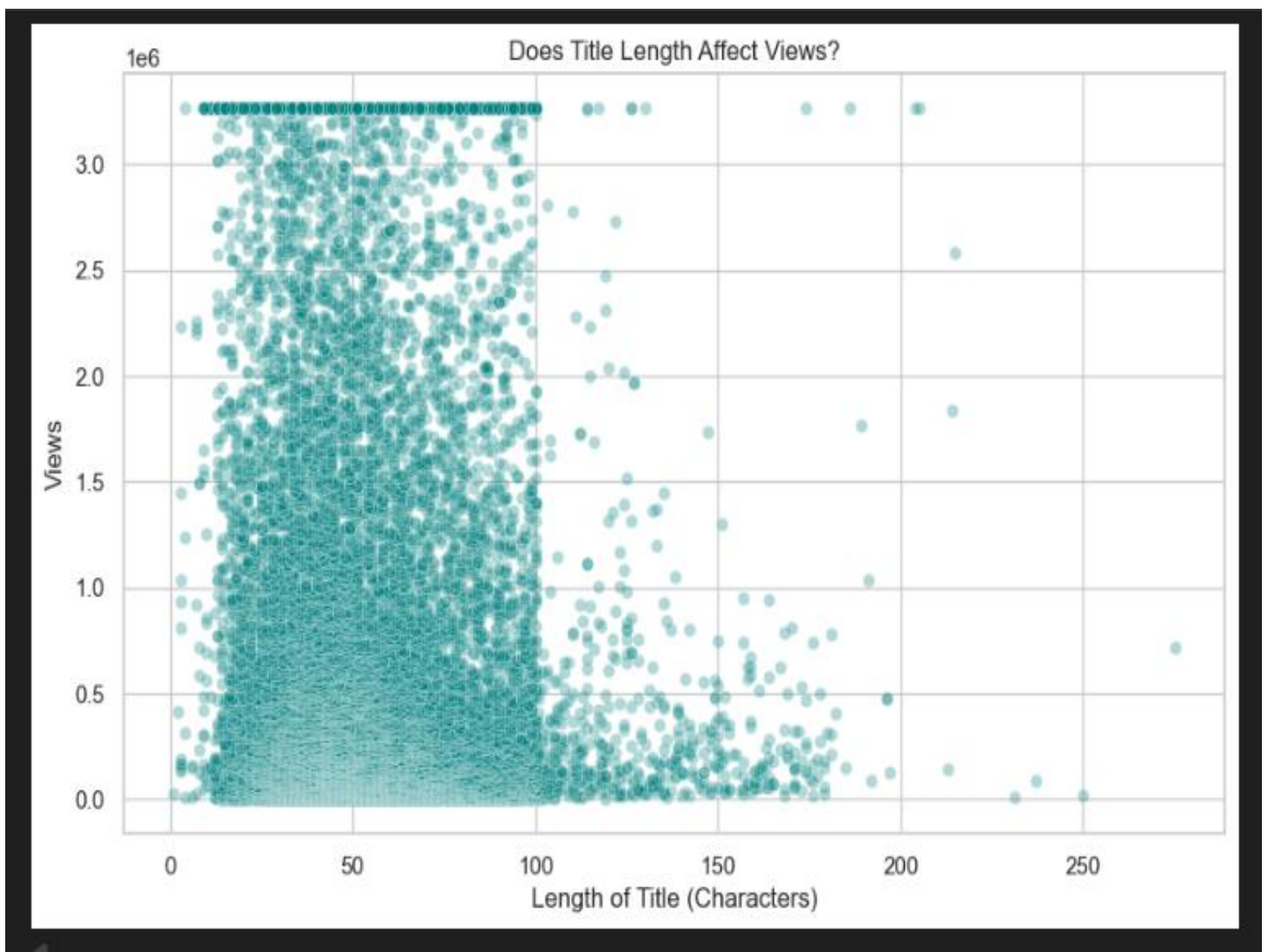
4. **Temporal Analysis:** The "Golden Window" By plotting the frequency of trending videos against publish_hour:

- Finding: There is a significant spike in videos published between 1:00 PM and 4:00 PM.
- Inference: Uploading during this window allows the video to be indexed and gain initial velocity just before the peak evening traffic (6 PM - 9 PM). Videos uploaded too late (after 8 PM) struggle to gain enough momentum to trend the next day.



5. Textual Analysis:

- Word Cloud: Visual analysis of titles emphasized words like "Official", "Trailer", "Video", "2024". This highlights the platform's bias toward "Official" studio content.
- Title Length: Trending videos overwhelmingly utilize concise titles (30-60 characters). Titles exceeding 100 characters showed a sharp drop in frequency, suggesting that users prefer quick, scannable information.



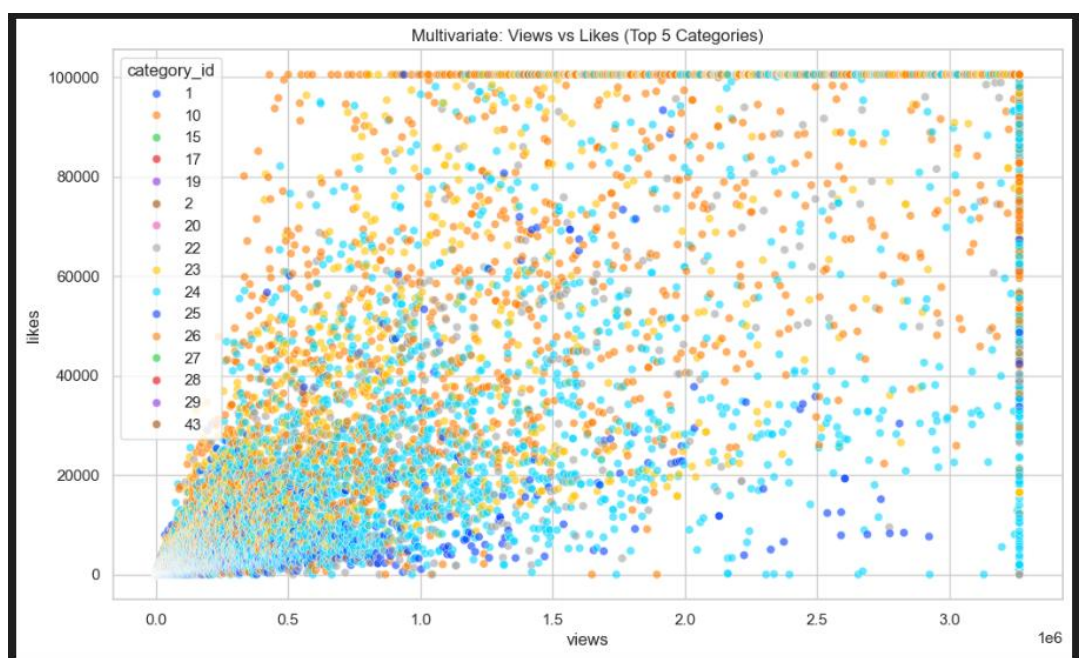
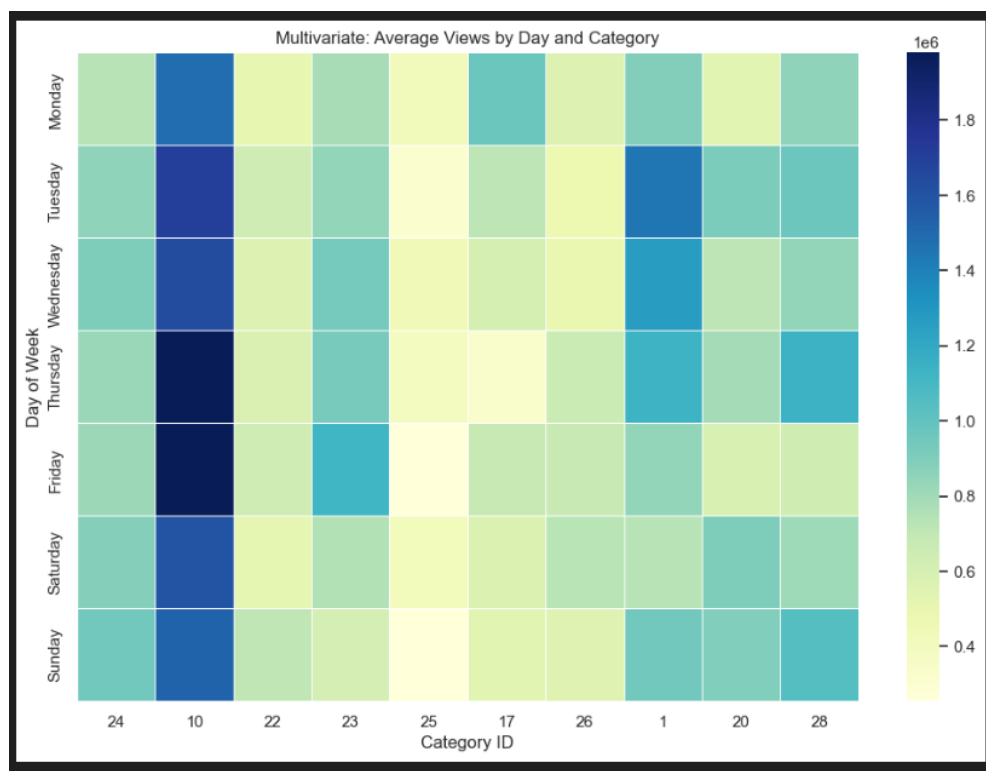
6. Multivariate Analysis:

- **Average Views by Day and Category:**

Certain categories receive higher views on specific days, with better performance observed during mid-week.

- **Views vs Likes:**

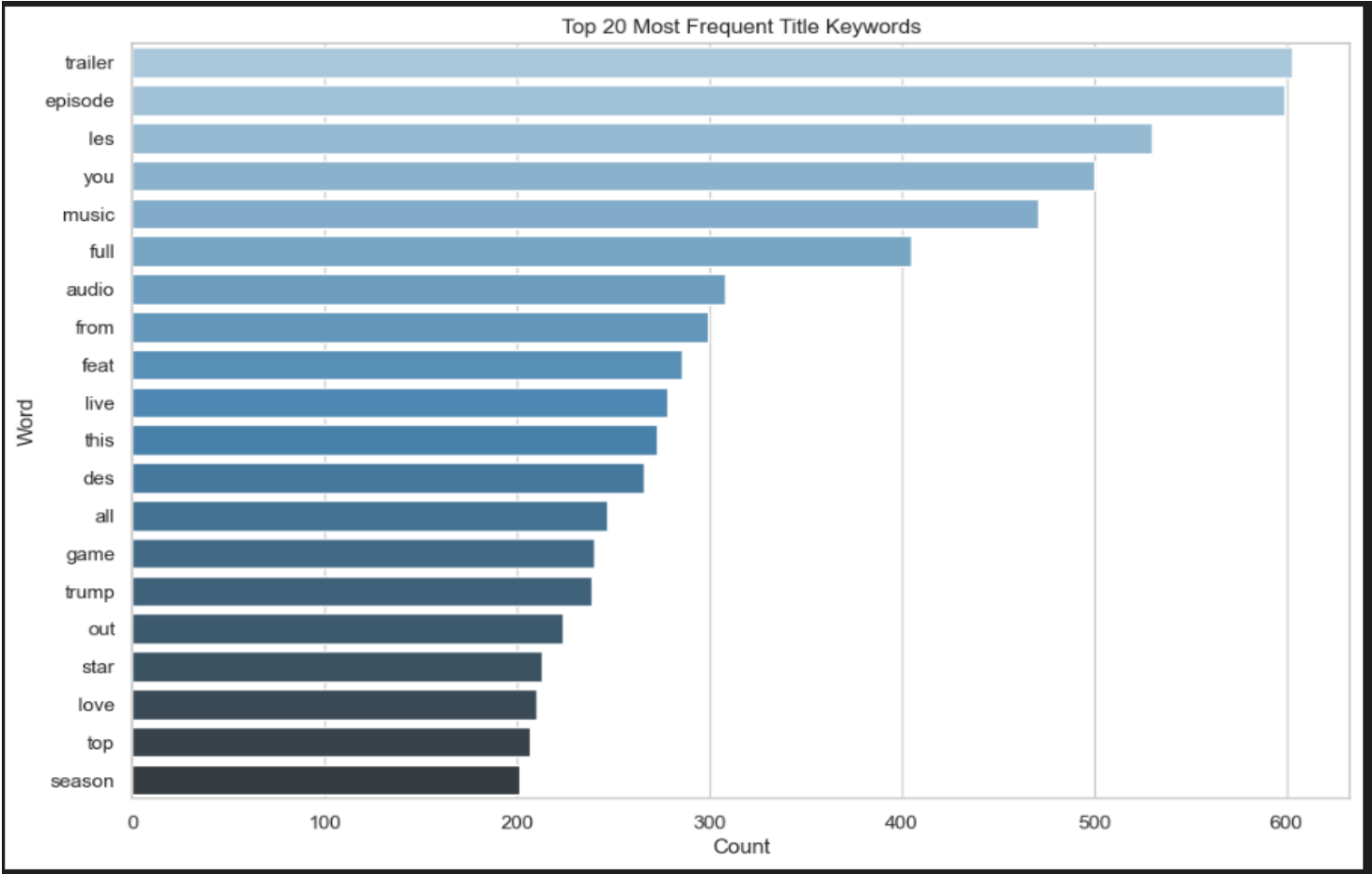
A positive correlation exists between views and likes, though engagement varies across categories.



Advance Insights:

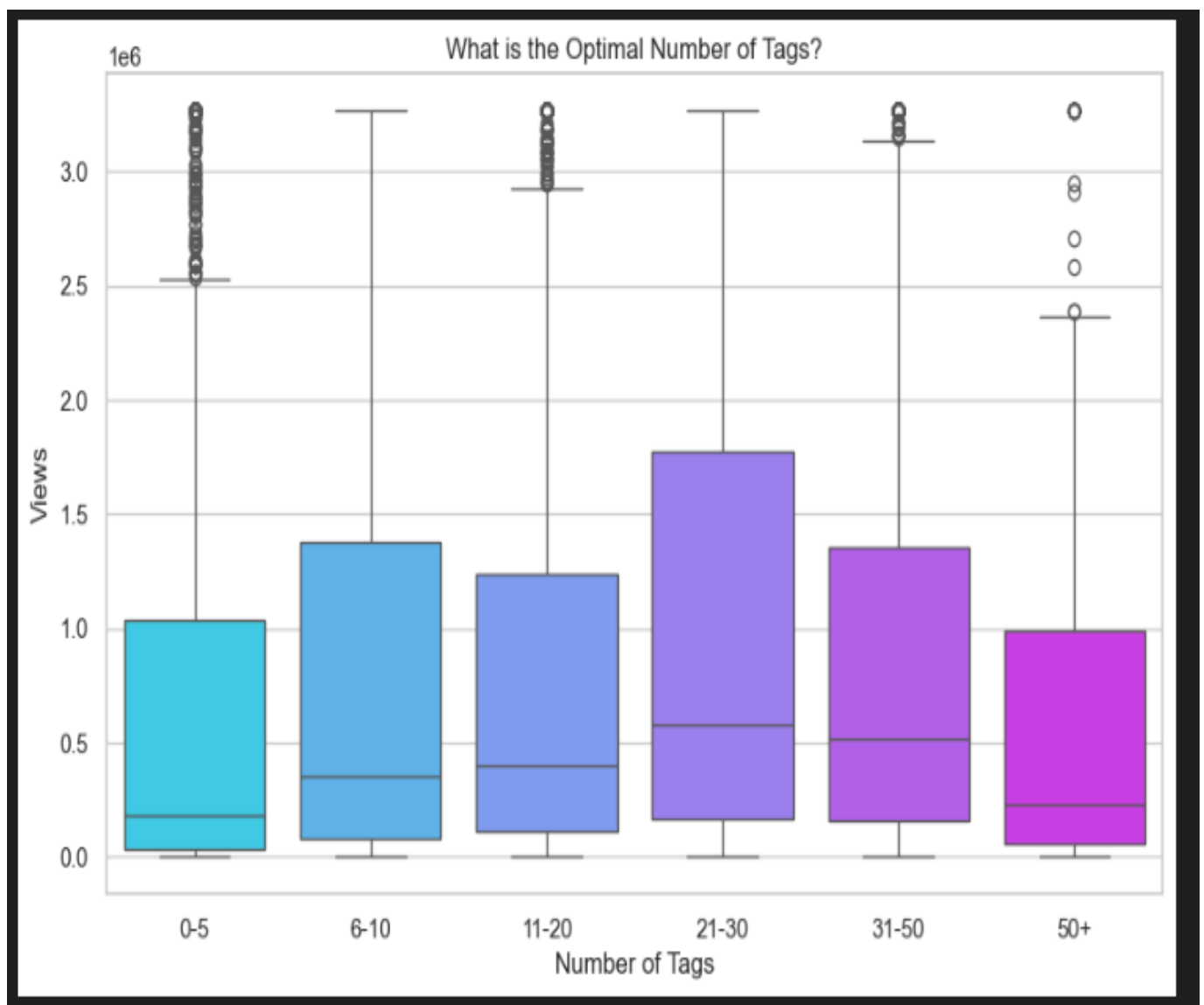
- **Top 20 Most Frequent Title Keywords**

This analysis highlights the most commonly used words in trending video titles. Keywords such as “trailer,” “episode,” “music,” and “full” appear most frequently, indicating a strong focus on entertainment and media-related content. The presence of generic terms like “you,” “this,” and “all” suggests the use of engagement-driven and click-optimized titles. Overall, title keywords reflect content strategies aimed at maximizing visibility and audience interest.



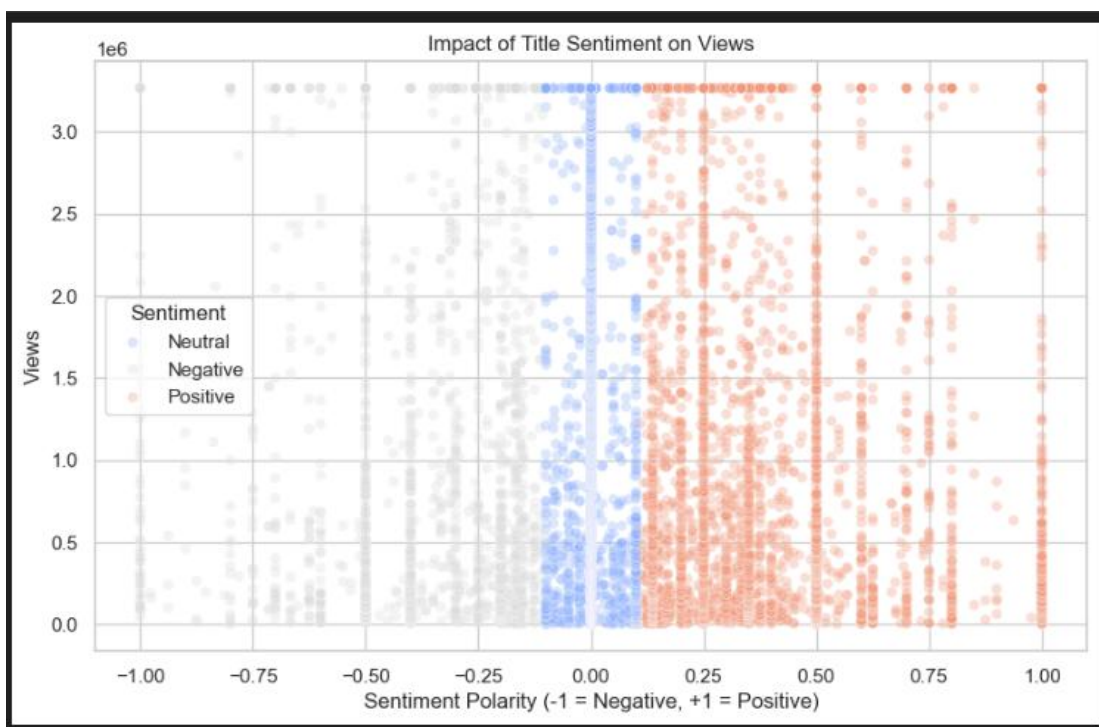
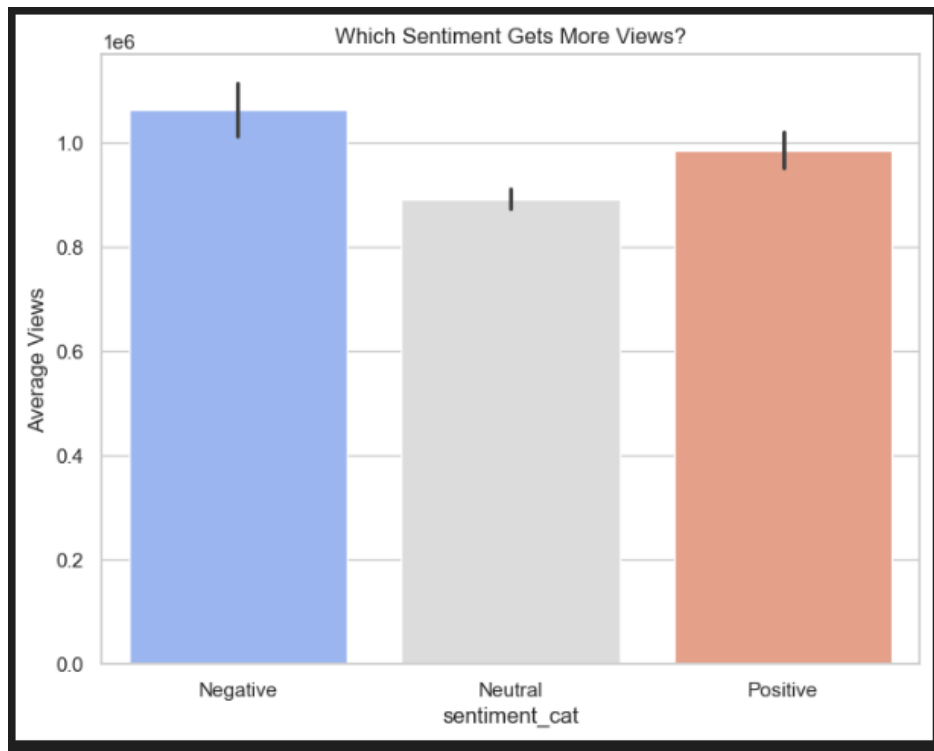
Optimal Number of Tags:

This analysis examines the relationship between the number of tags used and video views. Videos with moderate tag counts (around 21–30 tags) tend to achieve higher median views compared to very low or excessive tagging. Using too few or too many tags shows diminishing returns in visibility. This suggests that balanced and relevant tagging is more effective than excessive keyword usage.



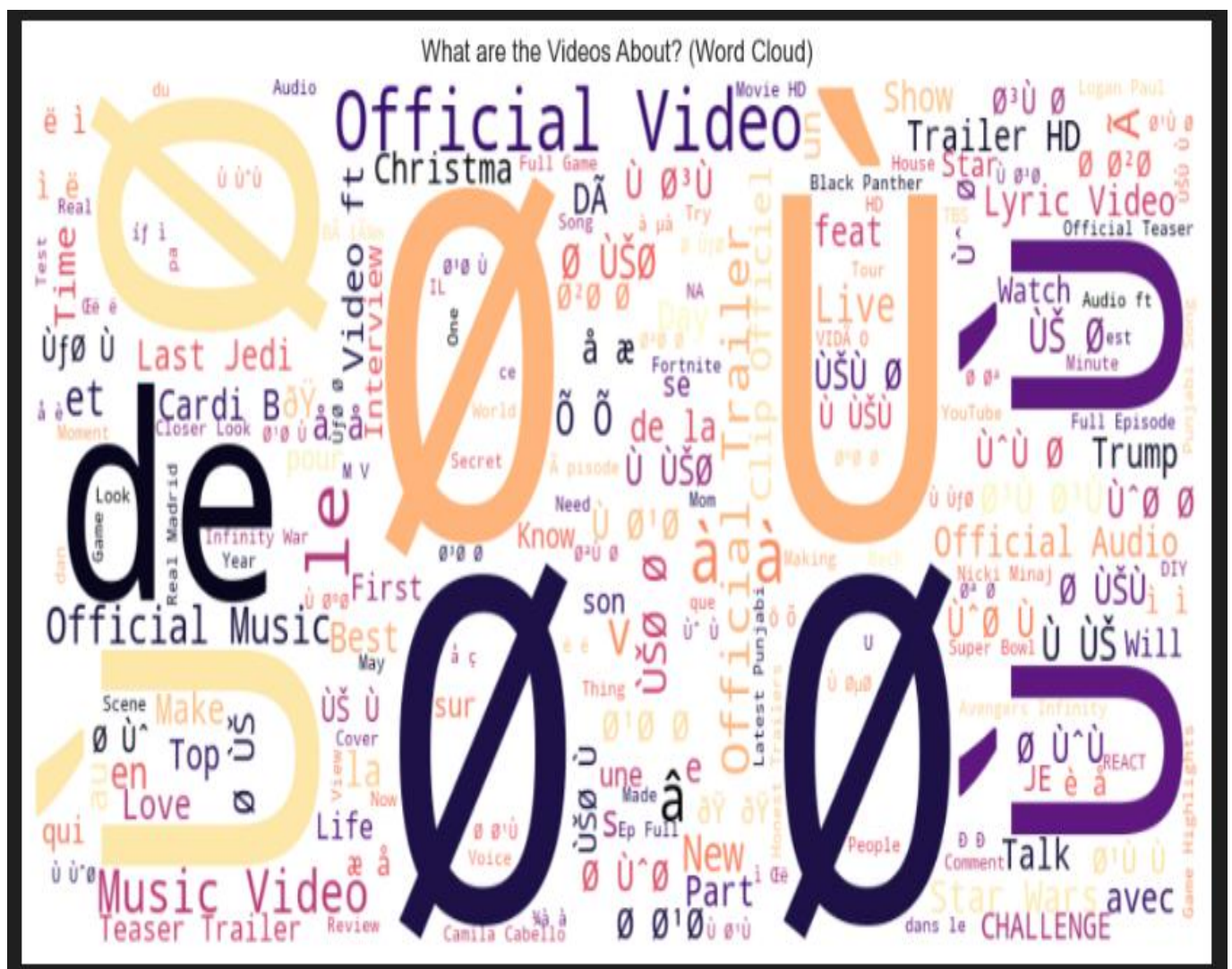
Sentiments

The sentiment analysis evaluates the impact of title polarity on video views. Videos with positive and negative sentiments tend to receive higher average views compared to neutral titles. Negative sentiment slightly outperforms others, indicating that emotionally charged titles attract more attention. Overall, sentiment plays a role in visibility, but it is not the sole factor influencing view counts.

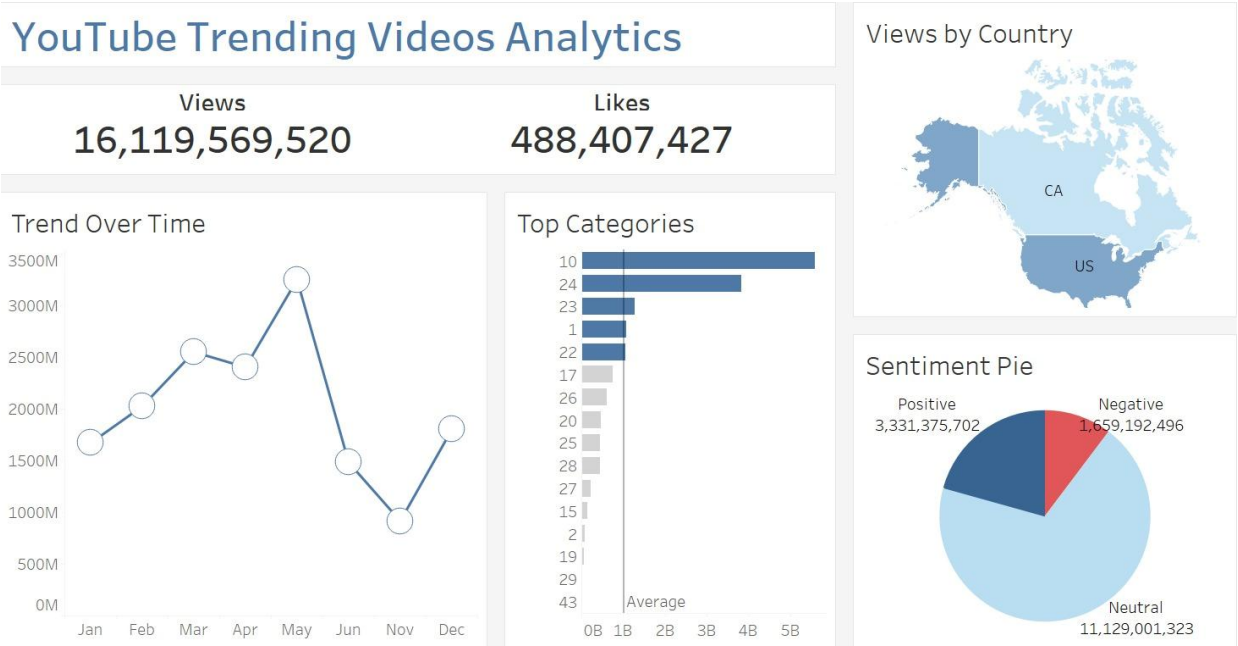


Highlights

The visualization shows the most frequently used words in YouTube video titles. Common terms such as “official,” “video,” “music,” “trailer,” and “audio” dominate, indicating a strong preference for standardized and recognizable title formats. These keywords reflect content strategies aimed at improving discoverability and audience engagement.



Dashboard :



This dashboard provides an exploratory data analysis (EDA) of YouTube trending videos, focusing on overall engagement, time-based trends, content categories, geographic distribution, and audience sentiment.

The dataset shows extremely high engagement, with 16.1 billion total views and 488 million likes, highlighting the massive reach of trending content on YouTube. These metrics indicate the strong influence of platform algorithms and viral content consumption.

The trend over time analysis reveals a steady increase in views from the beginning of the year, peaking around May, followed by a decline and partial recovery towards the end of the year. This pattern suggests seasonal variations in user activity and content consumption behavior.

The category-wise analysis shows that a few categories dominate total views, with category 10 leading significantly, followed by categories 24 and 23. This indicates a concentration of audience interest in specific types of content.

Geographically, the United States contributes the highest number of views, followed by Canada, emphasizing North America’s dominance in YouTube’s trending ecosystem.

The sentiment analysis indicates that most user interactions are neutral, followed by positive sentiment, while negative sentiment accounts for the smallest share. This suggests that trending videos generally receive widespread engagement without strong polarization.

FUTURE SCOPE

While this project provides valuable insights into the factors influencing YouTube video virality, there are several avenues for extending the analysis into a more robust and actionable content optimization framework. **Sentiment analysis of user comments** can be further developed using advanced NLP techniques to classify audience reactions into positive, negative, and neutral categories. Linking these sentiments with engagement metrics such as likes, comments, watch time, and trending duration can offer deeper insights into audience perception, emotional resonance, and long-term video performance.

Thumbnail and visual content analysis using computer vision methods can help identify influential visual elements, including the presence of faces, text overlays, colour contrast, brightness, and layout complexity. Understanding how these features affect click-through rates and initial viewer attention can assist creators in designing visually compelling thumbnails that improve discoverability and engagement.

In addition, **predictive modelling** using machine learning algorithms such as Random Forest and XGBoost can move the study from descriptive to predictive analytics. By leveraging early engagement indicators, these models can forecast a video's trending potential and overall reach, enabling creators to make data-driven decisions on content scheduling and promotion. Furthermore, **cross-regional analysis** can reveal cultural differences in content preferences and viewing patterns, while **multi-modal data integration** combining textual, visual, and engagement features can provide a more holistic understanding of virality drivers.

Finally, future work can focus on **real-time analytics and dashboard integration**, allowing creators and analysts to monitor video performance immediately after upload. By incorporating live engagement signals and automated alerts, such systems could support rapid content optimization, adaptive marketing strategies, and continuous feedback loops. This extension would transform the analysis from a static post-hoc study into a dynamic decision-support tool for content creators and digital marketers.

CONCLUSION

The *YouTube Trending Video Analysis* project clearly establishes that virality is not a product of randomness, but rather the result of optimizing a combination of measurable and actionable factors. Through a structured and comprehensive exploratory data analysis (EDA), the study revealed that engagement metrics, particularly likes and comments, are far more indicative of a video's potential to trend than passive indicators such as view counts alone. This emphasizes the importance of fostering active audience participation, rather than simply attracting large numbers of viewers.

The analysis further illuminated patterns in content performance across different categories. Entertainment and Music consistently dominate the trending tab due to their broad appeal, reflecting general audience preferences. However, niche categories such as Comedy, despite having smaller audiences, demonstrated higher engagement ratios, suggesting that focused, well-targeted content can outperform broader categories in terms of viewer interaction. This insight underscores the value of understanding both the size and responsiveness of target audiences when planning content strategy.

Temporal factors also play a significant role in influencing trending potential. Videos uploaded during late afternoon hours consistently showed higher probabilities of appearing on the trending list, highlighting the importance of strategic scheduling. This finding provides content creators with actionable guidance on when to publish, complementing efforts to optimize titles, descriptions, and metadata.

Beyond specific metrics, the project reinforces the broader principle that data-driven strategies can transform content creation from an intuitive, trial-and-error process into a structured, evidence-based approach. By systematically analysing video characteristics—including title length, tag density, category, and timing—creators can make informed decisions that maximize reach and engagement. Additionally, the project demonstrates how advanced data analytics techniques, such as correlation analysis, temporal modelling, and feature engineering, can provide deeper insights into platform algorithms that are otherwise opaque.

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